Algebra-Relative OAQEC/JLMS Reconstruction Does Not Certify Maximality

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Abstract

Let $\mathcal{M} \subseteq \mathcal{B}(\mathcal{H}_{\text{code}})$ be a finite-dimensional von Neumann algebra reconstructable on a boundary region or boundary algebra in the usual operator-algebra quantum-error-correction sense. For every $q \geq 2$, the finite central direct-sum extension

$$\mathcal{H}_{\text{code}}^{(q)} = \bigoplus_{j=1}^{q} \mathcal{H}_{\text{code}}, \qquad \mathcal{M}_q = \mathcal{M} \otimes \mathbb{C}^q \simeq \bigoplus_{j=1}^{q} \mathcal{M}$$

is reconstructable blockwise whenever the boundary representation and recovery are extended blockwise. A quotient-visible record obtained by composing with a central character $\pi_k : \mathbb{C}^q \to \mathbb{C}$ sees only the selected summand:

$$Q_k = \mathrm{id}_{\mathcal{M}} \otimes \pi_k : \mathcal{M}_q \to \mathcal{M}.$$

Thus OAQEC/JLMS data certify reconstruction of the supplied algebra. They do not, by themselves, certify that the supplied algebra is the full logical algebra against hidden finite central refinements. The obstruction disappears only after adding a maximality premise, complementary recovery/full-algebra identification, center-separating states or operations, or a physical rule excluding the finite center.

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1 Setup

Let $\mathcal{H}_{\text{code}}$ be finite-dimensional and let $\mathcal{M} \subseteq \mathcal{B}(\mathcal{H}_{\text{code}})$ be a finite-dimensional von Neumann algebra. The finite-dimensional structure theorem writes

$$\mathcal{M} \simeq \bigoplus_{\alpha} \big(\mathcal{B}(\mathcal{H}_{a_\alpha}) \otimes I_{\bar{a}_\alpha}\big).$$

In operator-algebra quantum error correction, a boundary region or boundary algebra reconstructs $\mathcal{M}$ when every $x \in \mathcal{M}$ has a boundary representative acting identically on the code, equivalently when the relevant erasure/recovery, commutant, and relative-entropy tests hold for the declared algebra.

Fix $q \geq 2$. Define

$$\mathcal{H}_{\text{code}}^{(q)} = \bigoplus_{j=1}^{q} \mathcal{H}_{\text{code}}, \qquad \mathcal{M}_q = \mathcal{M} \otimes \mathbb{C}^q \simeq \bigoplus_{j=1}^{q} \mathcal{M}.$$

Let p_j be the minimal central idempotents of $\mathbb{C}^q$. For a selected character $\pi_k : \mathbb{C}^q \to \mathbb{C}$, set

$$Q_k = \mathrm{id}_{\mathcal{M}} \otimes \pi_k : \mathcal{M}_q \to \mathcal{M}.$$

Theorem 1 (finite central direct-sum extension)**.** *Suppose $\mathcal{M}$ is reconstructable on a boundary region or algebra A in the finite-dimensional OAQEC sense. If the code, boundary representation, and recovery map are extended blockwise to $\mathcal{H}_{\text{code}}^{(q)}$, then $\mathcal{M}_q = \mathcal{M} \otimes \mathbb{C}^q$ is reconstructable blockwise on A. However, every declared reconstruction, recovery, modular, entropy, or JLMS record that factors through Q_k is insensitive to the nonselected central supports $\sum_{j \neq k} p_j$ and to the hidden central cardinality q.*

Proof. Let $R_A : \mathcal{M} \to \mathcal{B}(\mathcal{H}_A)$ denote a boundary representation of the declared algebra. Define

$$R_A^{(q)}(x_1, \ldots, x_q) = \bigoplus_{j=1}^{q} R_A(x_j)$$

on the blockwise boundary representation space. Since each component $R_A(x_j)$ acts on the corresponding copy of the code as x_j, the direct sum acts on $\mathcal{H}_{\text{code}}^{(q)}$ as $(x_1, \ldots, x_q)$. The same componentwise construction extends any recovery channel or algebra homomorphism used to state the OAQEC condition. Therefore $\mathcal{M}_q$ is reconstructable blockwise.

Now suppose a record factors through Q_k. If

$$(x_1, \ldots, x_q), (y_1, \ldots, y_q) \in \mathcal{M}_q \quad \text{with} \quad x_k = y_k,$$

then $Q_k(x_1, \ldots, x_q) = Q_k(y_1, \ldots, y_q)$. Consequently every boundary representative, recovery task, modular comparison, or relative entropy equality evaluated after the quotient has the same value for the two tuples, even if their nonselected components or central supports differ. The quotient-visible data therefore cannot determine the omitted finite central fiber or the number q. $\square$

2 Relation To JLMS

The JLMS statement compares boundary and bulk relative entropies for the bulk algebra or entanglement-wedge algebra under discussion. In the operator-algebra QEC formulation, the theorem is algebra-relative: the recovered object is the stated von Neumann algebra, possibly with center. The direct-sum extension above leaves that algebra-relative statement intact.

Proposition 1 (quotient-visible JLMS nonidentifiability). *Consider any family of code states, modular-flow tests, or relative-entropy comparisons whose declared bulk algebra is $Q_k(\mathcal{M}_q) \simeq \mathcal{M}$, or whose observed records factor through Q_k. The resulting JLMS/OAQEC data do not distinguish $\mathcal{M}$ from the selected quotient of $\mathcal{M} \otimes \mathbb{C}^q$, and hence do not certify that no hidden finite central direct-sum refinement was present.*

Proof. Relative entropy and modular data in the declared quotient are functions of the selected block. Changes to nonselected blocks change neither the quotient state nor the quotient algebra element. Thus the same quotient-visible JLMS/OAQEC record is compatible with multiple values of q and with multiple hidden central supports. □

Remark 1. If instead the state family includes all central sectors and their classical probabilities, full direct-sum relative entropy contains the usual classical relative-entropy term between central distributions. That is not a counterexample to the obstruction; it is a center-separating positive exit.

Proposition 2 (central-sector entropy is a positive exit). *Let ρ, σ be block-diagonal states on $\mathcal{M}_q \simeq \bigoplus_{j=1}^{q} \mathcal{M}$, written as*

$$\rho = \bigoplus_{j=1}^{q} p_j \rho_j, \qquad \sigma = \bigoplus_{j=1}^{q} r_j \sigma_j,$$

where p, r are probability vectors and ρ_j, σ_j are normalized states on the jth copy of $\mathcal{M}$. If $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$, then

$$S(\rho\|\sigma) = D(p\|r) + \sum_{j:p_j>0} p_j\, S(\rho_j\|\sigma_j),$$

where

$$D(p\|r) = \sum_{j:p_j>0} p_j \log \frac{p_j}{r_j}$$

is the classical relative entropy of the central-sector distributions. Consequently, a JLMS/OAQEC comparison that includes arbitrary central probability vectors does not factor through Q_k: it can detect changes in the finite center through the classical term $D(p\|r)$.

Proof. The logarithm of a block-diagonal positive operator is block diagonal on its support:

$$\log\rho = \bigoplus_{j:p_j>0} (\log p_j + \log\rho_j), \qquad \log\sigma = \bigoplus_{j:r_j>0} (\log r_j + \log\sigma_j).$$

Using $S(\rho\|\sigma) = \operatorname{Tr}\rho(\log\rho - \log\sigma)$, the trace over the direct sum separates into block traces:

$$S(\rho\|\sigma) = \sum_j p_j(\log p_j - \log r_j) + \sum_j p_j\ \operatorname{Tr}\rho_j(\log\rho_j - \log\sigma_j).$$

This is the stated formula. If the comparison class varies the central probability vector, the first term varies, so the record is no longer quotient-visible through a single selected summand. □

3 Positive Exits

The obstruction is intentionally conditional. It is killed, or demoted to a routine application, by any hypothesis that supplies finite-center accountability:

Exit premise	Effect
Full logical algebra stipulated	If $\mathcal{M}$ is defined to be the complete logical algebra of the code, then $\mathcal{M} \otimes \mathbb{C}^q$ is a different code algebra, not a hidden refinement of the same target.
Complementary recovery/maximality	If the theorem being invoked proves a unique maximal algebra from both region and complement data, then omitted central summands are excluded by hypothesis or theorem.
Center-separating states or operations	If the comparison includes states, operations, or boundary readouts that distinguish the central projections p_j, then the data no longer factor through Q_k. Proposition 2 is the relative-entropy version of this exit.
Physical exclusion	If the holographic model forbids such finite classical superselection labels, the extra premise should be named.

4 Referee Question

The classification question is precise:

> Does the standard OAQEC/JLMS hypothesis being used already imply that the supplied algebra is maximal as the full logical algebra, or that all finite central sectors are measured, included, or physically excluded? If so, name the theorem or axiom. If not, algebra-relative reconstruction does not by itself certify finite-center accountability.

The desired external response is not encouragement. It is one of: known finite-dimensional OAQEC fact, false statement, missing positive hypothesis, physically irrelevant comparison class, or useful expository no-go.

References

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